Vegetarian Dietary Patterns and Mortality in Adventist Health Study 2

Importance Some evidence suggests vegetarian dietary patterns may be associated with reduced mortality, but the relationship is not well established. Objective To evaluate the association between vegetarian dietary patterns and mortality. Design Prospective cohort study; mortality analysis by Cox proportional hazards regression, controlling for important demographic and lifestyle confounders. Setting Adventist Health Study 2 (AHS-2), a large North American cohort. Participants A total of 96 469 Seventh-day Adventist men and women recruited between 2002 and 2007, from which an analytic sample of 73 308 participants remained after exclusions. Exposures Diet was assessed at baseline by a quantitative food frequency questionnaire and categorized into 5 dietary patterns: nonvegetarian, semi-vegetarian, pesco-vegetarian, lacto-ovo–vegetarian, and vegan. Main Outcome and Measure The relationship between vegetarian dietary patterns and all-cause and cause-specific mortality; deaths through 2009 were identified from the National Death Index. Results There were 2570 deaths among 73 308 participants during a mean follow-up time of 5.79 years. The mortality rate was 6.05 (95% CI, 5.82–6.29) deaths per 1000 person-years. The adjusted hazard ratio (HR) for all-cause mortality in all vegetarians combined vs non-vegetarians was 0.88 (95% CI, 0.80–0.97). The adjusted HR for all-cause mortality in vegans was 0.85 (95% CI, 0.73–1.01); in lacto-ovo–vegetarians, 0.91 (95% CI, 0.82–1.00); in pesco-vegetarians, 0.81 (95% CI, 0.69–0.94); and in semi-vegetarians, 0.92 (95% CI, 0.75–1.13) compared with nonvegetarians. Significant associations with vegetarian diets were detected for cardiovascular mortality, noncardiovascular noncancer mortality, renal mortality, and endocrine mortality. Associations in men were larger and more often significant than were those in women. Conclusions and Relevance Vegetarian diets are associated with lower all-cause mortality and with some reductions in cause-specific mortality. Results appeared to be more robust in males. These favorable associations should be considered carefully by those offering dietary guidance.